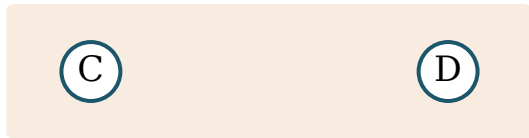


Optimal at 2 states: AC | DB



reassign mass  $w$



Optimal at 3 states: A | CD | B